HCL BigFix

Client Relevance Studio

User Guide

A modern, standalone IDE for writing, testing, and managing HCL BigFix Client Relevance expressions — with an integrated AI-powered assistant.

1. Getting Started

To ensure the underlying BigFix engine can access deep system metrics (such as local users, secure registry keys, or Windows Defender statuses), run the application with elevated privileges.

1. Download and extract the .zip file to a local folder on your Windows machine.
2. **BESClient installed:** The application automatically detects your local QnA engine path via the Windows registry. No additional steps required.
3. **BESClient NOT installed:** Download the standalone QnA utility from https://support.bigfix.com/bes/install/downloadbes.html and place qna.exe in the same directory as "BigFix Ops Studio.exe".
4. **Launch with elevation:** Right-click "BigFix Ops Studio.exe" and select "Run as administrator".

💡 Tip — First Run

On first launch the application window may take a few seconds to appear while the local HTTP server initialises on port 5055.

2. The Interface at a Glance

The Studio is divided into three resizable panes. Drag the horizontal or vertical dividers between panes to customise the layout.

Feature	Description
Left Sidebar	Your repository. Contains the built-in Relevance Library, your custom My Bookmarks, and your Recent History. Fully searchable.
Expression Editor	A CodeMirror-powered text area (top right) for writing and editing relevance. Features line numbers and bracket matching.
Evaluation Results	The results pane (bottom right). Displays parsed answers, execution time in milliseconds, and errors returned by the QnA engine.
Context Selector	A dropdown in the top toolbar to switch between Client Relevance, Session Relevance, and Action Script contexts.
AI Assistant	A slide-in panel supporting multiple AI providers. Generates expressions, explains syntax, and loads code directly into the editor.

3. Working with Contexts

BigFix Ops Studio now supports three evaluation contexts. Use the dropdown at the top-left of the toolbar to switch between them:

- Client Relevance — The default mode for querying endpoint properties.
- Session Relevance — For session-based queries used in the BigFix Console.
- Action Script — For writing and referencing BigFix Action Script snippets.

Each context maintains its own library, bookmarks, and history. Switching contexts automatically refreshes the sidebar.

4. Evaluating Relevance

Single Expression

Type your relevance query into the Expression Editor, then click the blue Evaluate button or press Ctrl + Enter.

Batch Evaluation

Write each expression on its own line. The Studio evaluates all of them in a single pass and stacks result blocks in the Results pane.

```
names of drives
operating system
free space of drive "C:"
uptime of operating system
```

⌨ Keyboard Shortcut

Ctrl + Enter evaluates from anywhere in the editor without reaching for the mouse.

5. Using the Built-In Library

The sidebar contains a categorised library of standard BigFix relevance snippets covering Operating System, Hardware, Network, Security, Services, and more. The library is context-aware — it updates automatically when you switch between Client Relevance, Session Relevance, or Action Script.

- **Browse:** Click Expand All or Collapse All to navigate categories.
- **Load:** Click any item card to instantly load that snippet into the Expression Editor.
- **Search:** Use the "Search expressions..." bar to filter the library, bookmarks, and history simultaneously.
- **SmartSense Explain:** Click the lightbulb icon on any library card to send that expression directly to the AI Assistant for an explanation.

6. Saving Custom Bookmarks

Bookmarks are context-specific — a bookmark saved in Client Relevance mode only appears in that context.

1. Type or paste your expression into the Expression Editor.
2. Click the Bookmark button in the Expression Editor header.
3. Enter a short title and an optional description when prompted.
4. Your snippet is saved under My Bookmarks in the sidebar.

Deleting a Bookmark

Hover over the bookmark in the sidebar and click the Trash Can icon that appears on the right.

7. Navigating Your History

Every successful Evaluate action automatically saves the query to the Recent History accordion in the sidebar. History is also context-specific.

- Stores your last 100 queries per context.
- Queries older than 10 days are automatically purged.
- Click any history item to reload it into the editor.
- Click Clear History inside the accordion to wipe the log for the current context only.

8. Searching and Filtering Results

When a query returns many answers, use the live filter to narrow them down.

- Type in the "Search answers..." box in the Evaluation Results header.
- Result blocks are filtered in real-time, showing only matching rows.
- Matched text is highlighted in yellow for quick scanning.

9. Exporting Your Data

Share results with colleagues or attach them to tickets in six formats.

1. Click the Export button in the Evaluation Results header.
2. Select your format: CSV, PDF, HTML, TXT, JSON, or XML.
3. Choose your scope: Search Results (matching rows only) or All Results (everything).
4. A save dialog prompts you to choose a destination on your machine.

10. AI Assistant (Multi-Provider)

BigFix Ops Studio now features a fully upgraded AI Assistant with a multi-provider, cascading fallback architecture. You can choose between three AI providers — Groq, Google Gemini, or Anthropic Claude — based on your organisation's data policies and API access.

10.1 How the AI Architecture Works

Within each provider, the system maintains a priority list of models. Every chat request tries the most capable model first. If that model hits a rate limit (HTTP 429), the system silently falls back to the next model in the list and retries — keeping the assistant functional even under heavy usage.

ℹ Why Cascading Fallback?

Free-tier AI endpoints impose strict Tokens-Per-Minute (TPM) limits. Because each request injects the BigFix Relevance Library as context, token consumption per request is high. Cascading to a faster, higher-limit model ensures the assistant stays responsive rather than locking you out.

10.2 Provider Comparison

Choose the provider that best fits your needs or compliance requirements:

Provider	Best Free Model	Free Limits	How to Get Key
Groq	`llama-3.3-70b-versatile`	30 RPM / 12K–30K TPM	console.groq.com → API Keys
Google Gemini	`gemini-2.5-flash`	15 RPM / 1M TPM	aistudio.google.com → Get API Key
Anthropic Claude	`claude-3-5-sonnet-latest`	~5 RPM / ~10K TPM*	console.anthropic.com → Claim $5 free credits

** Anthropic's free tier is a one-time $5 evaluation credit. A funded billing account is required once the credit is exhausted.*

10.3 Obtaining API Keys

Groq (Recommended for speed)

1. Go to console.groq.com and sign in.
2. Click API Keys in the left sidebar.
3. Click Create API Key and copy the generated key.

Google Gemini (Recommended for generous free quota)

1. Go to aistudio.google.com.
2. Click Get API Key in the left sidebar.
3. Click Create API key in new project and copy the key.

Anthropic Claude (Best reasoning quality)

1. Go to console.anthropic.com.
2. Claim the $5 free evaluation credit.
3. Navigate to API Keys and create a key.

10.4 Activating Your Provider

Open a Command Prompt in the application folder and run the appropriate command:

Start with Groq:

```
"BigFix Ops Studio.exe" --provider=groq --groq-api-key="YOUR_KEY_HERE"
```

Start with Gemini:

```
"BigFix Ops Studio.exe" --provider=gemini --gemini-api-key="YOUR_KEY_HERE"
```

Start with Claude:

```
"BigFix Ops Studio.exe" --provider=claude --claude-api-key="YOUR_KEY_HERE"
```

💾 Key Persistence
Your API key and provider preference are saved automatically to the Windows Registry. On all future launches, open the application normally — no extra flag needed.
To switch providers, re-run the application with --provider= and the new provider name.
To replace an expired key, re-run with the appropriate --xxx-api-key= flag.

10.5 Opening the Assistant

- Click the AI Assistant button (labelled BETA) in the top-right toolbar.
- The assistant panel slides in from the right side of the screen.
- Click ✖ or click AI Assistant again to close it.

10.6 What It Can Do

Feature	Description
Write Relevance	Describe what you want in plain English and the assistant generates a ready-to-use relevance expression.
Explain Expressions	Paste an expression and ask what it returns, line by line. Use the SmartSense lightbulb button for instant explanation.
Debug Errors	Paste an error message from the Results pane and ask for a fix.
Suggest Alternatives	Ask for a more efficient or cross-platform version of an existing expression.
Answer Concepts	Ask general BigFix questions such as the difference between property types or inspector namespaces.
Context-Aware Help	The assistant knows which context you are in (Client Relevance, Session Relevance, or Action Script) and tailors its answers accordingly.

10.7 SmartSense AI Explanations

The SmartSense feature provides instant, inline explanations directly from library items or your current editor content.

- **Lightbulb button on library cards:** Click to send that expression to the AI Assistant for an explanation without loading it into the editor.
- **SmartSense button in editor header:** Click to explain either your current selection or the entire editor content.
- **Auto-hint:** Typing up the inspectors in the editor triggers SmartSense autocomplete suggestions for common relevance properties.

10.8 Loading Code into the Editor

Any relevance code block in an assistant response displays a "Click to load in editor" label. Clicking it instantly places the expression into the Expression Editor — ready to evaluate without copy-pasting.

✅ Best Practice

After loading AI-generated code, always click Evaluate to verify the result on your machine before deploying to endpoints.

10.9 Example Prompts

```
relevance for checking if Windows Defender real-time protection is enabled
relevance to list all installed patches with their KB numbers
explain what this does: (name of it, value of it) of values of key
"HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion" of registry
relevance for CPU architecture, socket count, and core count
```

10.10 Privacy & Data

🔒 Note

Your chat messages are sent to the selected AI provider's API for processing. Do not include sensitive credentials or personally identifiable information in chat messages.

Your API keys are stored locally on your machine and are never transmitted outside the respective provider's own API.

11. Quick Pro-Tips

Feature	Description
Fast Copying	Click any Q: or A: row in the Results pane to copy that text to your clipboard. The row flashes blue to confirm.
Persistence	Bookmarks, history, and the current editor content are saved to local cache. Closing and reopening restores everything exactly as you left it.
Clear Editor	Click the Clear button next to Evaluate to wipe the editor and results board.
Resize Panes	Drag the dividers between panes to customise your layout at any time.
Ctrl + Enter	Evaluate expressions without leaving the keyboard.
Batch + Filter	Evaluate 10+ expressions at once, then use Search answers to drill into a specific result.
Context Switching	Switch between Client Relevance, Session Relevance, and Action Script — each has its own library, bookmarks, and history.
Provider Switching	Switch AI providers at any time by restarting with --provider=groq/gemini/claude.

12. Troubleshooting

Feature	Description
Engine not found	Ensure qna.exe is in the same directory as "BigFix Ops Studio.exe", or that BESClient is installed and registered in the Windows registry.
Port 5055 in use	Another process is using port 5055. Close it, or edit the PORT variable in config.py.
AI Assistant disabled	The API key for the selected provider is missing or invalid. Re-run the application with the appropriate --xxx-api-key= flag to save a new key.
Rate limit errors	All models for the selected provider are rate-limited. Wait 60 seconds and retry, or switch to a provider with a higher free quota (Gemini offers 1M TPM).
AI response garbled	Try rephrasing your prompt or restarting the assistant panel.
Results pane empty	If the expression evaluates but returns no rows, the QnA engine returned no answers — not an error. Check your expression logic.
Want to switch provider	Run: "BigFix Ops Studio.exe" --provider=groq (or gemini / claude). The new preference is saved for future launches.